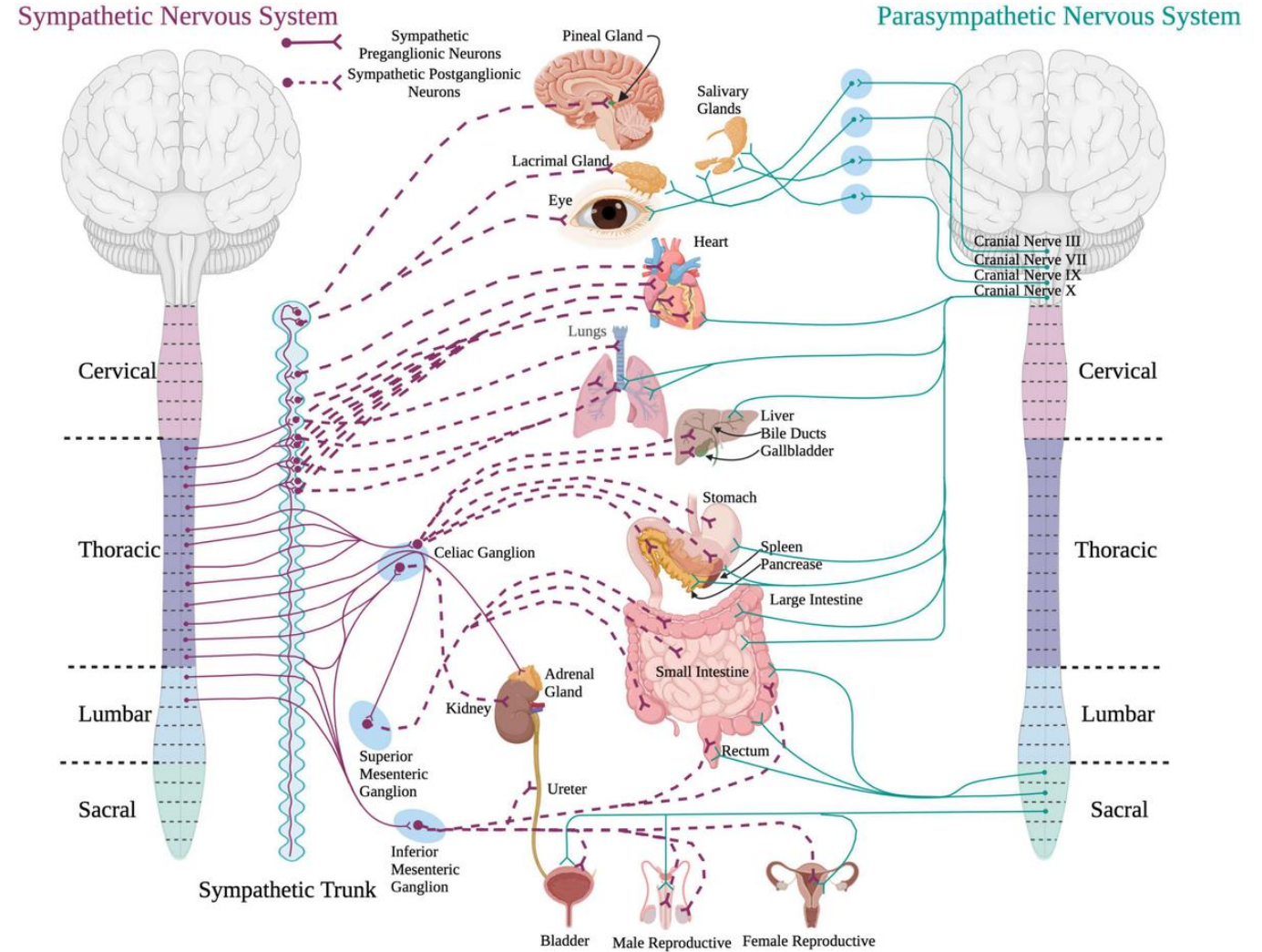


Autonomic Nervous System II

BIOL2251

Chapter 16

Sections 16.7 – 16.10



Lecture Quiz



Test what you know!



10 questions (MC)



Quiz automatically submits at 2:30 PM



Password = cortex

Make sure you sign in before the end of lecture!



Dual Innervation and Visceral Reflexes

Dual Innervation and Visceral Reflexes

- **Instructions**

- Take out a piece of paper and a pen/pencil. Briefly answer the following questions:

1. **Describe what dual innervation means and explain the concept of autonomic tone.**

- Why is autonomic tone vital for homeostasis?

2. **Draw a basic flow chart of a visceral reflex arc.**

- Make sure to identify the main difference between a long reflex and a short reflex.

- **Bloom's taxonomy = Remember/Understand**

Dual Innervation and Visceral Reflexes

- **Describe what dual innervation means and explain the concept of autonomic tone. Why is autonomic tone vital for homeostasis?**
 - **Dual Innervation:** Most visceral organs receive nerves from both the sympathetic and parasympathetic divisions of the ANS, where each division has opposing effects
 - Sympathetic = increases heart rate
 - Parasympathetic = decreases heart rate
 - **Autonomic Tone:** Background level of activity in an autonomic motor neuron
 - Vital for homeostasis because it allows an organ to increase or decrease its activity instantly, rather than having to turn a system completely 'off' before adjusting

Sympathetic-Only Targets

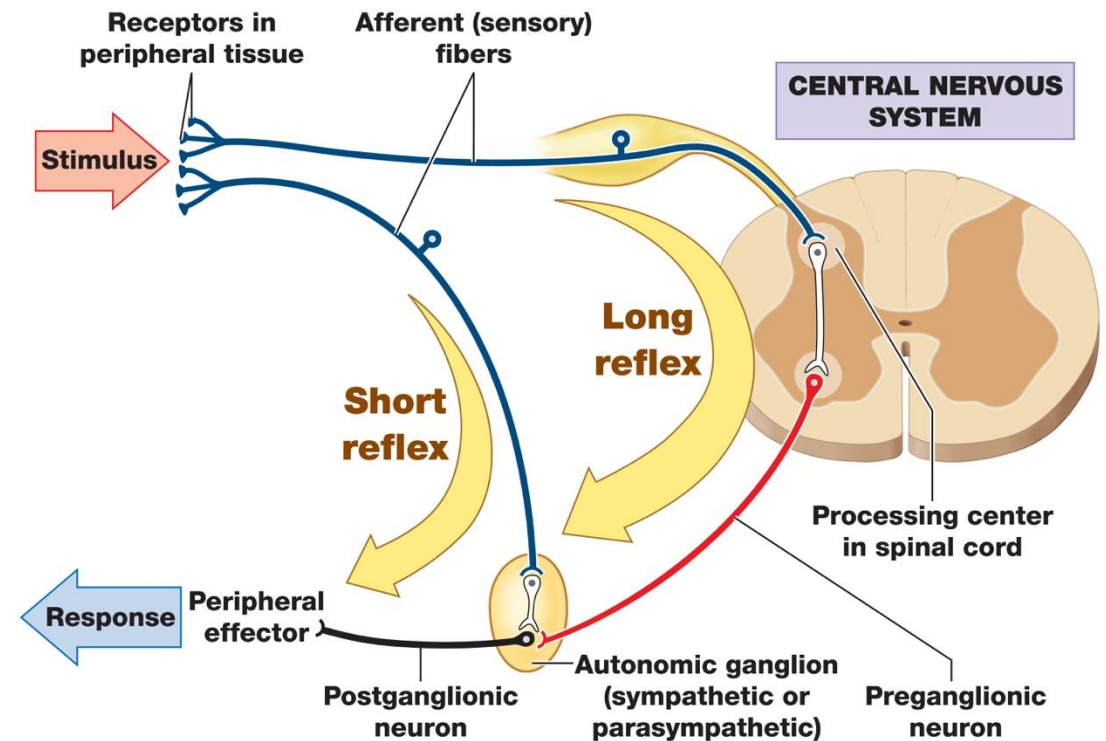
- **List at least 3 organs or structures that are only innervated by the sympathetic division**
 - What is the typical effect of sympathetic activation on these three structures?

- Sweat Glands = Increased secretion
- Blood Vessels = Dilation or constriction
- Adrenal Glands = Secretion of epinephrine and norepinephrine
- Posterior Pituitary Gland = Secretion of ADH
- Mucous Glands (Respiratory System) = Mucus secretion
- Skeletal Muscles = Increased force of contraction, glycogen breakdown
- Adipose Tissue = Release of fatty acids
- Arrector Pili Muscles = Erection of hairs

Dual Innervation and Visceral Reflexes

- **Flow Chart**

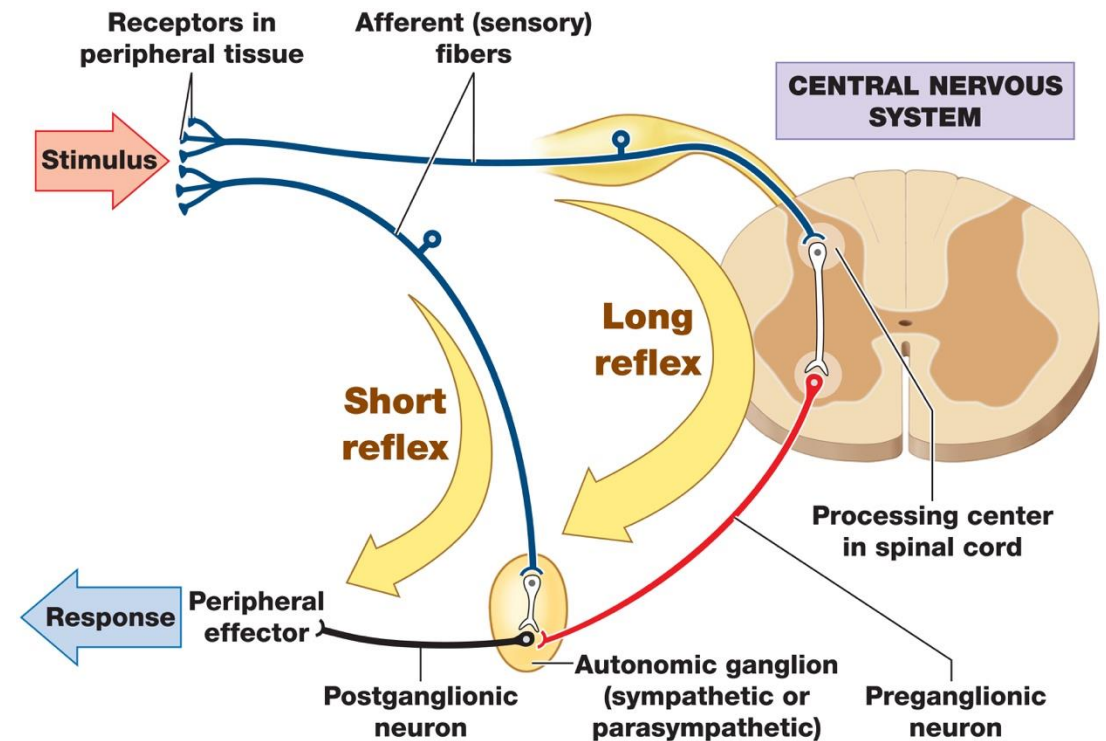
1. Receptor
2. Sensory Neuron
3. Processing Center
 - Preganglionic/Ganglionic and Postganglionic
4. Motor Neurons
5. Peripheral Effectors

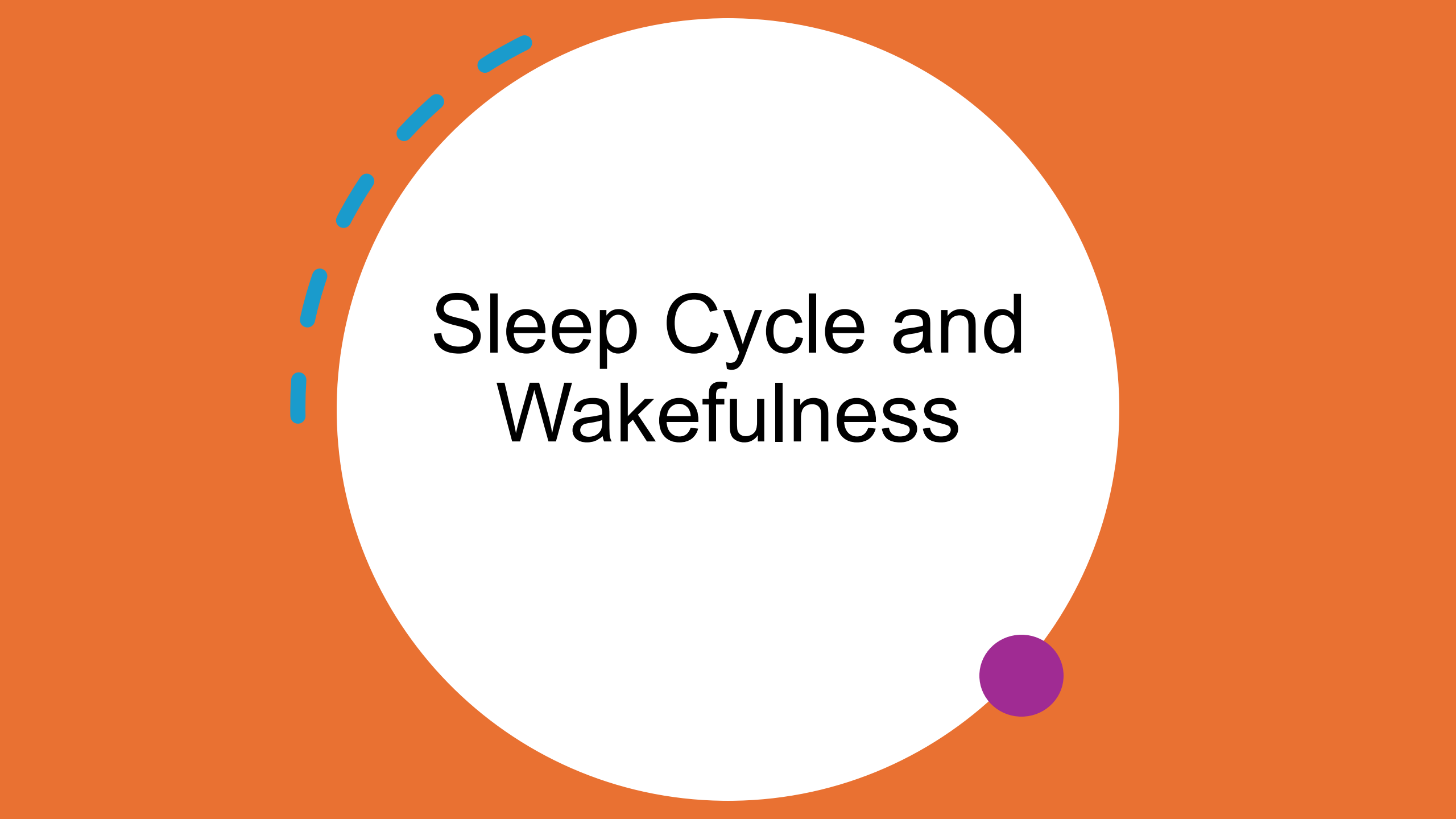


Dual Innervation and Visceral Reflexes

- **Long vs. Short Reflexes**

- Long reflexes involve the CNS as the processing center
- Short reflexes bypass the CNS
 - Processing center is an autonomic ganglion





Sleep Cycle and Wakefulness

Sleep Cycle and Wakefulness

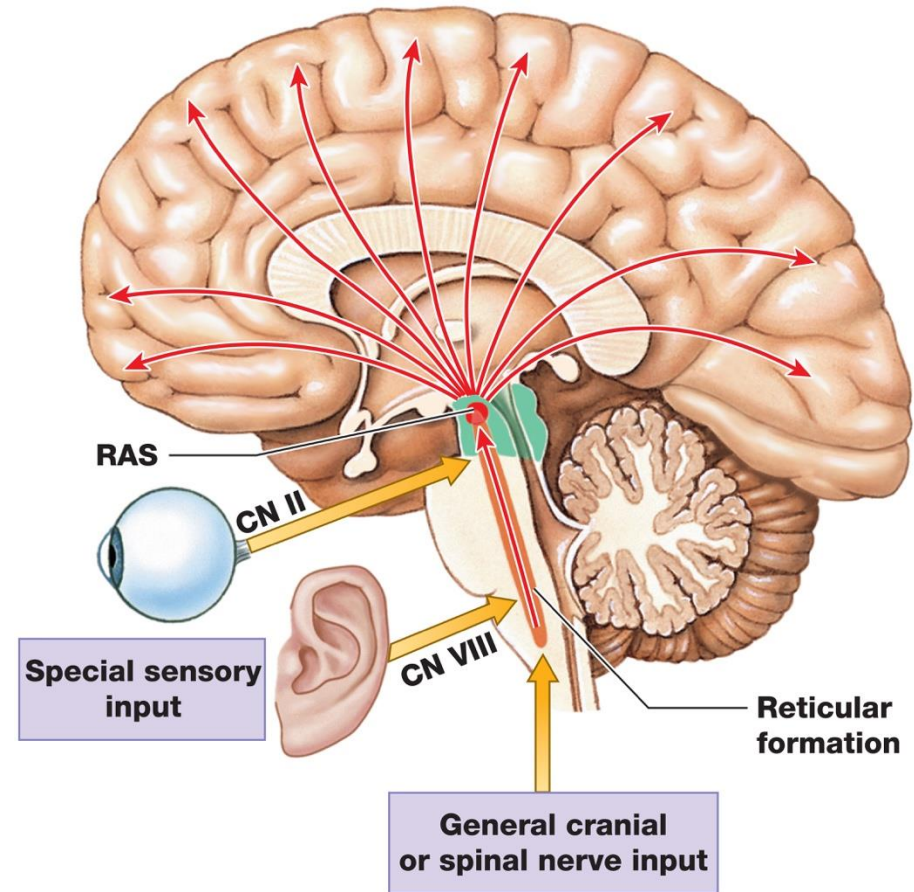
- **Instructions**

- Work with the person next to you. Create a two-part diagram to answer the following:
- **RAS and Wakefulness:** Create a simple diagram illustrating the flow of the Reticular Activating System (RAS). Your diagram should:
 - Start with sensory input
 - Show how the RAS maintains the awake state
- **Caffeine's Cheat Code:** Explain how caffeine affects the sleep-wake cycle based on your knowledge of the RAS and neurotransmitters.
 - Which natural molecule does it block?
- **Bloom's taxonomy = Understand/Apply**

Sleep Cycle and Wakefulness

RAS and Wakefulness

1. Sensory Input (via spinal cord + cranial nerves)
2. Reticular Formation (brainstem)
3. RAS (projects to thalamus, cerebral cortex)
4. Generalized Excitatory Effect on Cerebral Cortex
5. Conscious/Alert State Maintained



Sleep Cycle and Wakefulness

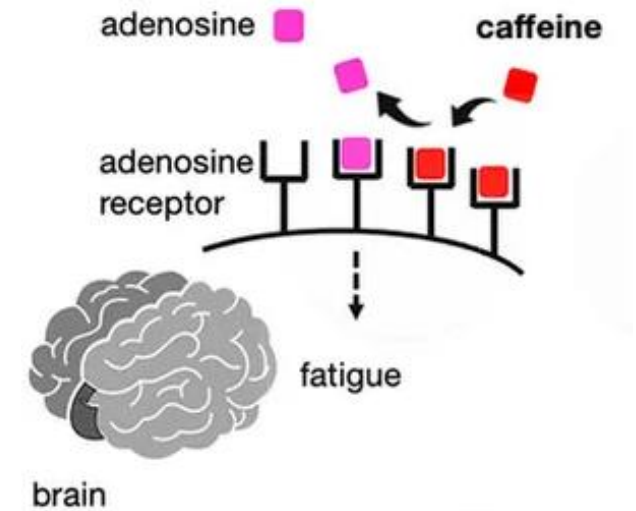
- **Explain how caffeine affects the sleep-wake cycle based on your knowledge of the RAS and neurotransmitters. Which natural molecule does it block?**
 - Caffeine blocks adenosine receptors
 - Adenosine is an inhibitory molecule that builds up during the day and promotes sleepiness
 - By blocking the adenosine receptors, caffeine prevents this 'sleep signal' from reaching the brain

Sleep Cycle and Wakefulness

- Norepinephrine = promotes wakefulness
- Adenosine = promotes sleep
 - **Blocked by caffeine!**

Wake-promoting neurotransmitters	NREM	REM
Glutamate	GABA	Acetylcholine
Acetylcholine	Galanin	Glutamate
Dopamine	Adenosine	GABA
Norepinephrine	Melatonin	Glycine (muscle atonia)
Serotonin		
Histamine		
Orexin/hypocretin		

Note: NREM = nonrapid eye movement; REM = rapid eye movement;
GABA = gamma-amino-butyric acid.



When adenosine binds to its receptor, this causes fatigue. Caffeine prevents adenosine binding and thereby delays fatigue

Too much caffeine

- Nervousness/jitters
- Nausea
- Elevated heart rate
- Anxiety
- Headache
- Dysphoria
- Insomnia

Memory Consolidation



Memory Consolidation

- **Instructions**

- Work in small groups (~4-6 per whiteboard)
 - Be ready to hold up your whiteboard to share your answers!
- **Memory Differentiation: Create a quick table or list that differentiates between:**
 - Fact memories vs. skill memories
 - Short-term memories vs. long-term memories
- **Consolidation at the Cellular Level: Explain the four cellular mechanisms that help with memory consolidation**
 - Pathway must explicitly **state the two key brain structures that are essential for memory consolidation to occur**

- **Bloom's taxonomy = TBD**

Memory Consolidation

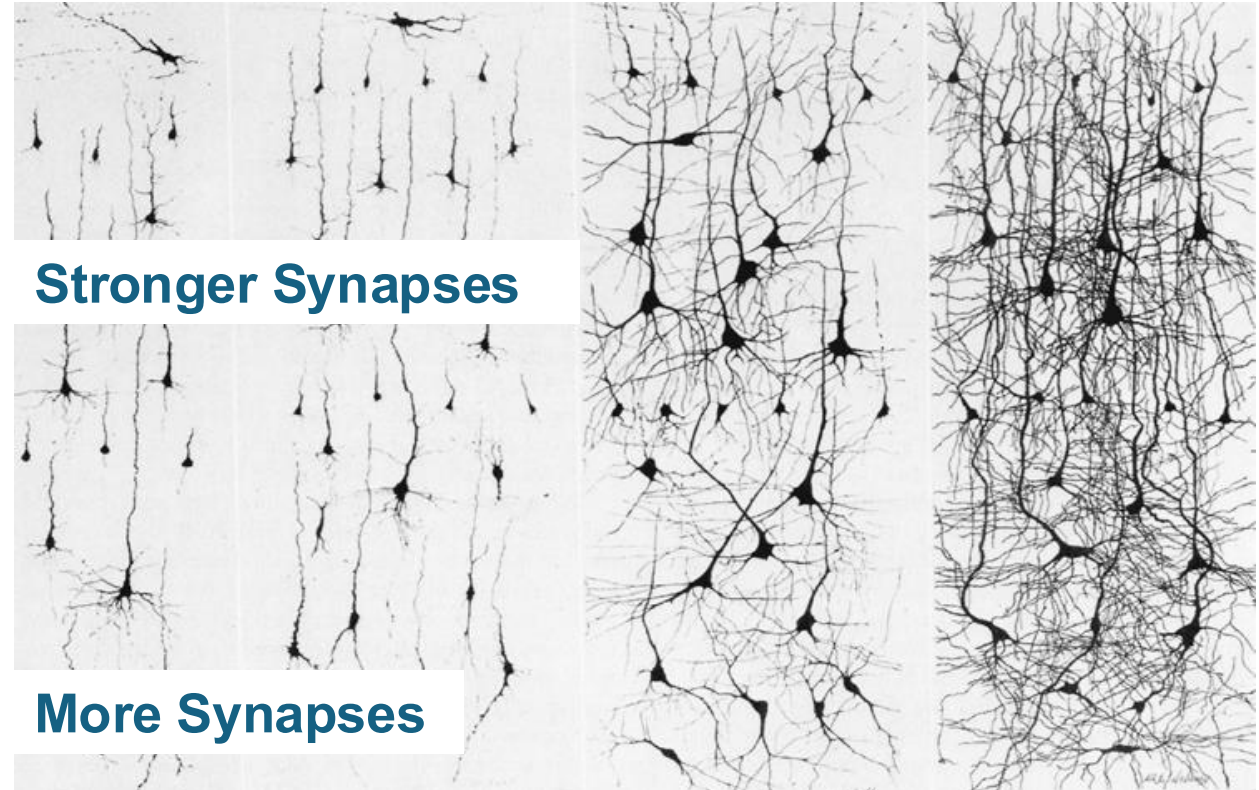
Memory Differentiation

- **Fact Memories:** Specific information (names, dates, concepts); requires conscious recall; often learned quickly
- **Skill Memories:** Learned motor behaviors and unconscious actions (riding a bike, muscle memory); gained by repetition; relatively permanent
- **Short-Term Memories:** Last seconds to minutes; limited capacity (a few pieces of info); often based on patterns of neuronal firing
- **Long-Term Memories:** Last days, weeks, or a lifetime; virtually limitless capacity; **established by structural changes at synapses (consolidation)**

Memory Consolidation

Cellular Mechanisms

- **Increased NT release** at specific synapses
- **Increased receptors** on the post-synaptic membrane
- Growth of **new axon collaterals and synaptic terminals**
- Growth of **new dendrites and dendritic spines**



Where are memories stored?

Association areas (e.g., visual, auditory) and the frontal lobe

Memories can be linked across association areas = stronger memory



Dopamine, Instant Gratification, and Delayed Rewards

Dopamine, Instant Gratification, and Delayed Rewards

- **Instructions**

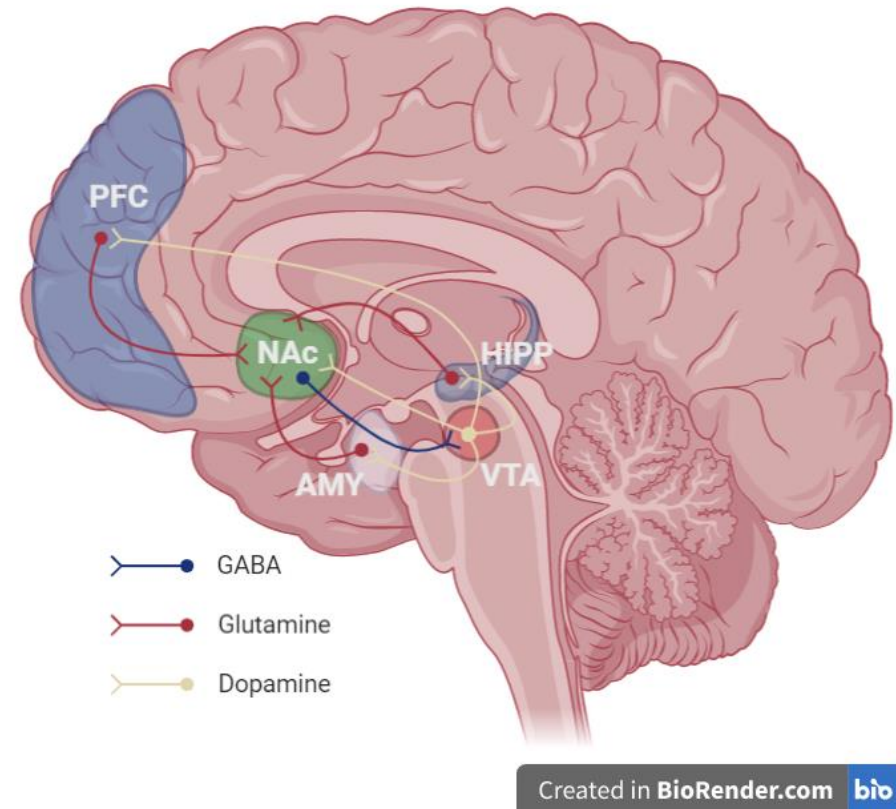
- Whole class discussion
- **Reward Pathway:** We will work through the main pathway of the dopamine reward circuit, including the two main structures involved
- We will then use this knowledge to work through two problems
- **Bloom's taxonomy = Analyze/Evaluate**

Dopamine Reward Circuit

↑ Dopamine

↓ Serotonin

- **Ventral tegmental area (VTA)** = origin of the circuit; primary production center for dopamine
- **Nucleus accumbens (NAc)** = primary target of the VTA; releases a surge of dopamine; controls motor function
- **Hippocampus (HIPP)** = formation of memories
- **Amygdala (AMY)** = emotions
- **Prefrontal cortex (PFC)** = attention and planning



Dopamine, Instant Gratification, and Delayed Rewards

- **Instructions**

- You will use your knowledge of the dopamine reward circuit to address the following questions:
 - **Instant Gratification:** How do common sources of instant gratification influence the dopamine receptors compared to natural rewards?
 - **Gratification Gap:** Explain why it is becoming increasingly difficult for the brain to maintain the motivation to work toward goals that require delayed gratification
- **Bloom's taxonomy = Analyze/Evaluate**

Dopamine, Instant Gratification, and Delayed Rewards

- **How do common sources of instant gratification influence the dopamine receptors compared to natural rewards?**
 - Instant gratification sources provide a rapid, unpredictable, and concentrated surge of dopamine
 - Unpredictability mimics the reward structure of addictive substances/behaviors, leading to compulsive seeking and rapid habit formation
 - Natural rewards are slower and release lower levels of dopamine
 - Modern instant gratification conditions the brain to expect the high, fast dopamine release, making natural rewards feel dull by comparison

Dopamine, Instant Gratification, and Delayed Rewards

- **Explain why it is becoming increasingly difficult for the brain to maintain the motivation to work toward goals that require delayed gratification**
 - Constant, unnaturally high surges of dopamine from instant gratification lead the brain to downregulate its dopamine receptors or reduce its baseline dopamine release
 - When the brain returns to the "effortful" activity, which provides a smaller, slower reward (delayed gratification), the activity feels less satisfying and requires more conscious effort to initiate
 - The brain learns that effort is associated with a low reward compared to the zero-effort, high-reward activities of instant gratification